

Digital Forensics

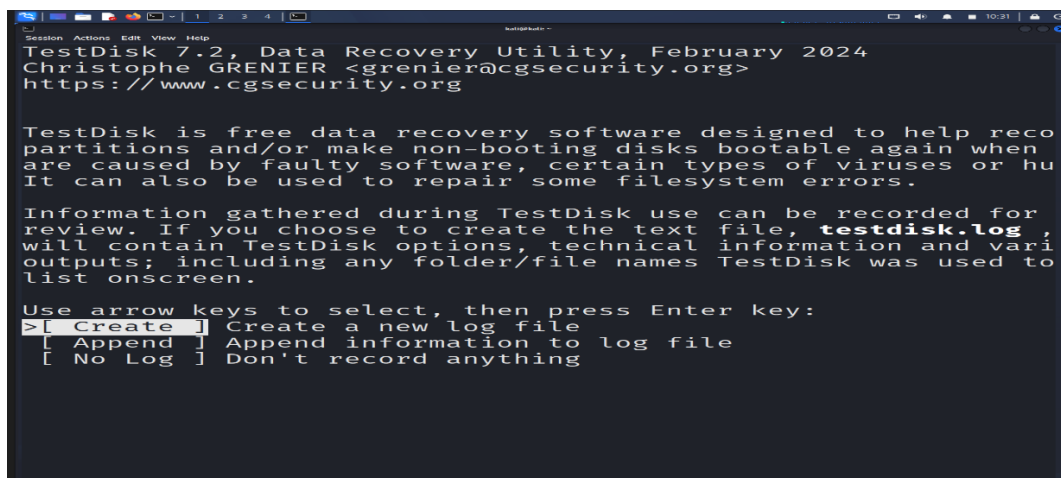
Ex.No:2 Recover deleted or damaged files from a storage device using Test Disk

Date:

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation

A screenshot of the TestDisk 7.2 terminal window. The window title is "TestDisk 7.2, Data Recovery Utility, February 2024". The user is identified as "Christophe GRENIER <grenier@cgsecurity.org>" with the website "https://www.cgsecurity.org". The text explains that TestDisk is free data recovery software designed to help recover partitions and/or make non-booting disks bootable again when caused by faulty software, certain types of viruses or hardware failures. It can also be used to repair some filesystem errors. It mentions that information gathered during use can be recorded for review in a text file named "testdisk.log", which will contain TestDisk options, technical information and various outputs, including any folder/file names TestDisk was used to list onscreen. It then prompts the user to use arrow keys to select and press Enter. The options are: "[Create] Create a new log file", "[Append] Append information to log file", and "[No Log] Don't record anything". The "Create" option is highlighted with a cursor.

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help reco
partitions and/or make non-booting disks bootable again when
are caused by faulty software, certain types of viruses or hu
It can also be used to repair some filesystem errors.

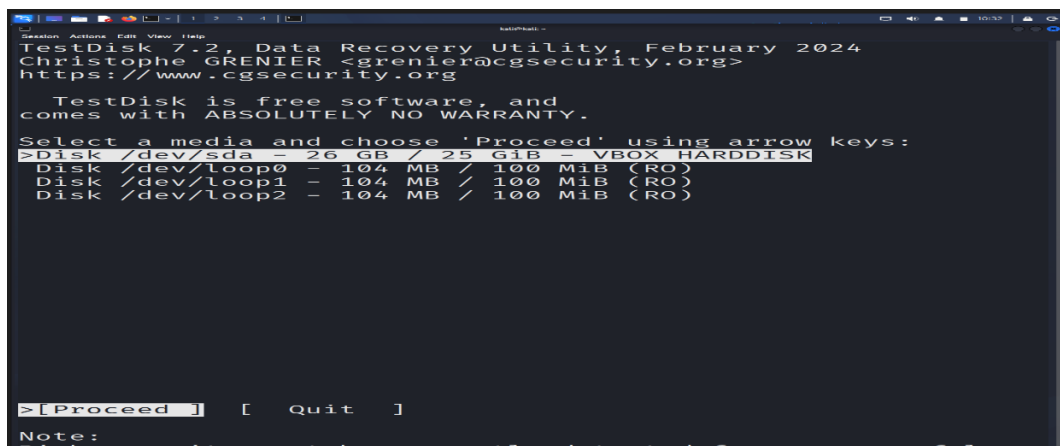
Information gathered during TestDisk use can be recorded for
review. If you choose to create the text file, testdisk.log ,
will contain TestDisk options, technical information and vari
outputs; including any folder/file names TestDisk was used to
list onscreen.

Use arrow keys to select, then press Enter key:
>[ Create ] Create a new log file
[ Append ] Append information to log file
[ No Log ] Don't record anything
```

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or
- you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk selection

All hard drives should be detected and listed with the correct size by TestDisk:

A screenshot of the TestDisk 7.2 terminal window. The window title is "TestDisk 7.2, Data Recovery Utility, February 2024". The user is identified as "Christophe GRENIER <grenier@cgsecurity.org>" with the website "https://www.cgsecurity.org". The text explains that TestDisk is free software, and comes with ABSOLUTELY NO WARRANTY. It prompts the user to select a media and choose 'Proceed' using arrow keys. The options are: ">Disk /dev/sda - 26 GB / 25 GiB - VBOX HARDDISK", "Disk /dev/loop0 - 104 MB / 100 MiB (RO)", "Disk /dev/loop1 - 104 MB / 100 MiB (RO)", and "Disk /dev/loop2 - 104 MB / 100 MiB (RO)". The "Disk /dev/sda" option is highlighted with a cursor. At the bottom, it prompts the user to press Enter to proceed or Quit.

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TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk /dev/sda - 26 GB / 25 GiB - VBOX HARDDISK
Disk /dev/loop0 - 104 MB / 100 MiB (RO)
Disk /dev/loop1 - 104 MB / 100 MiB (RO)
Disk /dev/loop2 - 104 MB / 100 MiB (RO)

>[ Proceed ] [ Quit ]

Note:
disk capacity must be correctly detected for a successful rec
```

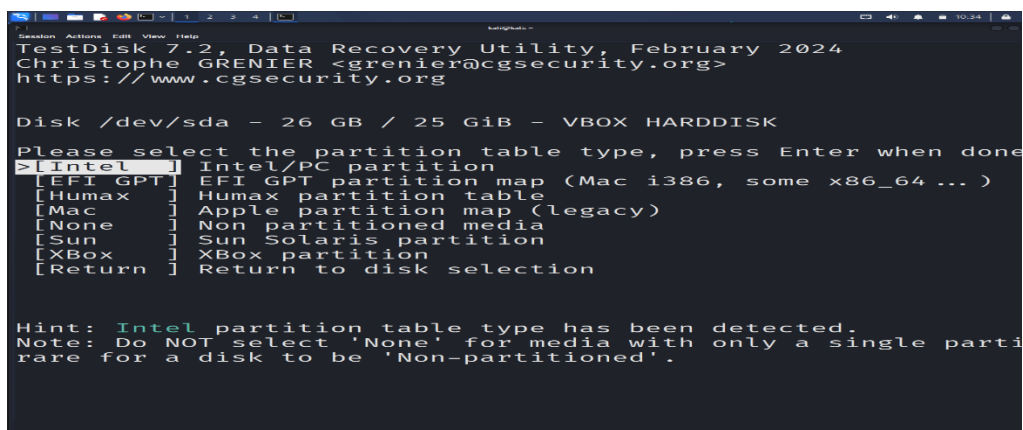
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- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

If available, use raw device /dev/rdisk* instead of /dev/disk* for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.



```
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Disk /dev/sda - 26 GB / 25 GiB - VBOX HARDDISK

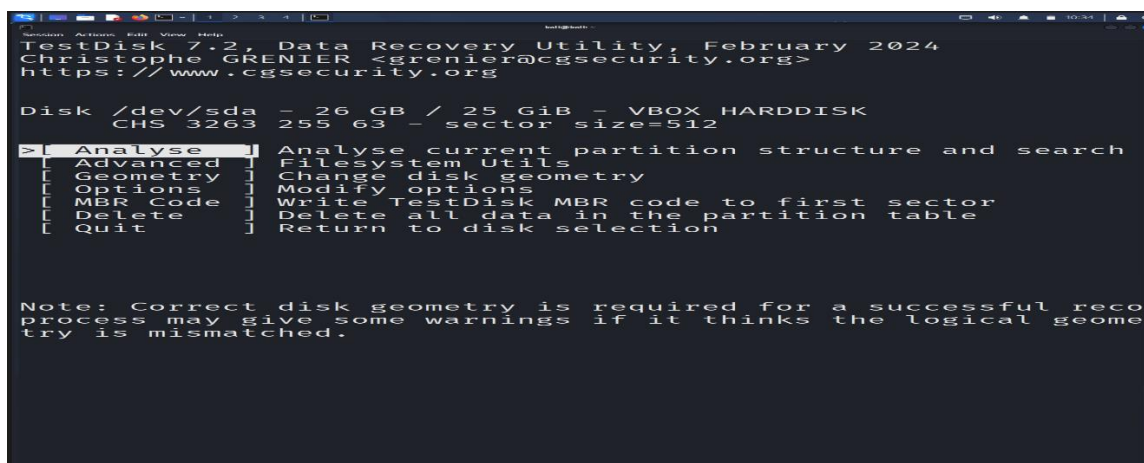
Please select the partition table type, press Enter when done
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64 ...)
[Humax] Humax partition table
[Mac] Apple partition map (legacy)
[None] Non partitioned media
[Sun] Sun Solaris partition
[XBox] XBox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single parti
rare for a disk to be 'Non-partitioned'.
```

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current partition table status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).



```
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Disk /dev/sda - 26 GB / 25 GiB - VBOX HARDDISK
CHS 3263 255 63 - sector size=512

>[Analyze] Analyze current partition structure and search
[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[MBR Code] Write TestDisk MBR code to first sector
[Delete] Delete all data in the partition table
[Quit] Return to disk selection

Note: Correct disk geometry is required for a successful reco
process may give some warnings if it thinks the logical geome
try is mismatched.
```

- Use the default menu "Analyze" to check your current partition structure and search for lost partitions.
- Confirm at Analyze with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.

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```
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Disk /dev/sda - 26 GB / 25 GiB - CHS 3263 255 63
Current partition structure:
Partition              Start          End          Size in s
ecto
1 * Linux              0  32 33    3090    9 55    4963942
Bad relative sector.
2 E extended LBA      3090    42 23    3263   105 42    278323
5 L Linux Swap        3090    42 25    3263   105 42    278323
2

*=Primary bootable  P=Primary  L=Logical  E=Extended  D=Delet
>[Quick Search]  [ Backup ]  Try to locate partition
```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for partitions

TestDisk displays the first results in real time.  (click on thumb to display the image).

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled **Partition 3**.

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Disk /dev/sda - 26 GB / 25 GiB - CHS 3263 255 63
Partition              Start          End          Size in sect
>* Linux              0  32 33    3090    9 55    49639424
P Linux Swap        3090    42 25    3263   105 42    2783232

Structure: Ok.  Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics
*=Primary bootable  P=Primary  L=Logical  E=Extended  D=Delet
Keys A: add partition, L: load backup, T: change type, P: list
t fileEnter: to continue
ext4 blocksize=4096 Large_file Sparse_SB Recover, 25 GB / 23
```

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- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the partition table or search for more partitions?

- **When all partitions are available** and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu Extd Part gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- **Since a partition, the first one, is still missing**, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.

A partition is still missing: Deeper Search

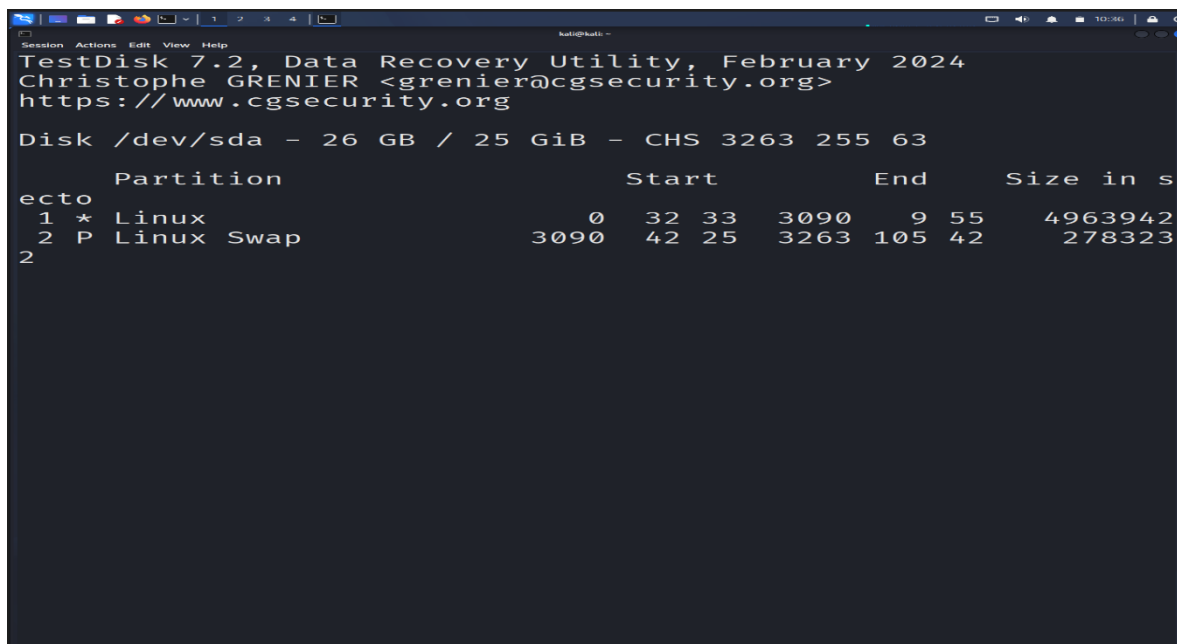
Deeper Search will also search for FAT32 backup boot sector, NTFS backup boot superblock, ext2/ext3 backup superblock to detect more partitions,

it will scan each cylinder  (click on thumb).

After the Deeper Search, the results are displayed as follows:

The first partition "**Partition 1**" was found by using backup boot sector. In the last line of your display, you can read the message "**NTFS found using backup sector!**" and the size of your partition. The "partition 2" is displayed twice with different size.

Partitions listed as D(eleted) will not be recovered if you let them listed as deleted. Both partitions are listed with status **D** for deleted, because they overlap each other. You need to identify which partition to recover.



```
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Disk /dev/sda - 26 GB / 25 GiB - CHS 3263 255 63

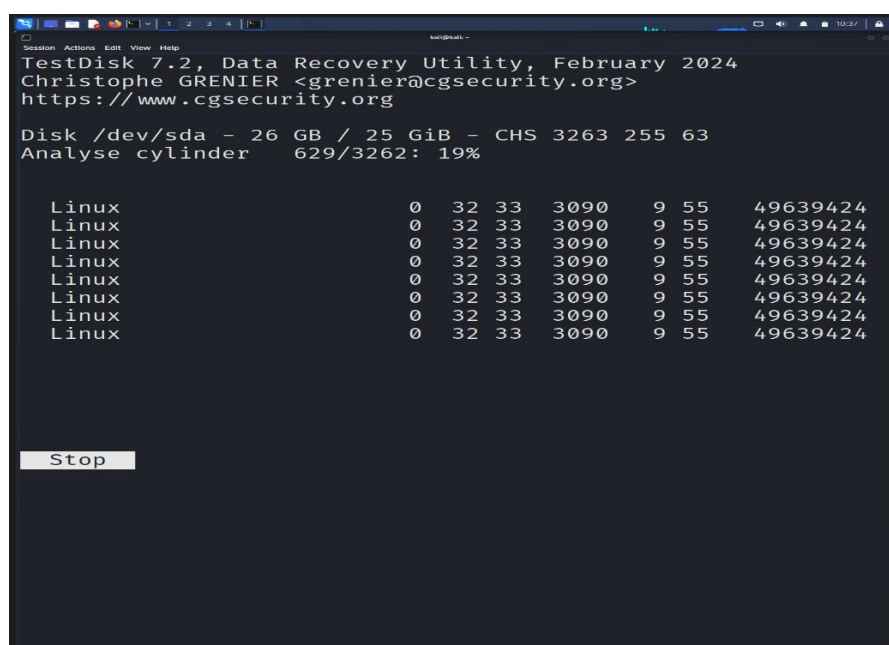
Partition      Start      End      Size in s
1 * Linux      0 32 33   3090 9 55   4963942
2 P Linux Swap 3090 42 25   3263 105 42  278323
```

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- Highlight the first partition Partition 2 and press **p** to list its data.

The file system of the upper logical partition (label Partition 2) is damaged  (click on thumb).

- Press q for Quit to go back to the previous display.
- Let this partition Partition 2 with a damaged file system marked as D(deleted).
- Highlight the second partition Partition 2 below
- Press p to list its files.



```
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Disk /dev/sda - 26 GB / 25 GiB - CHS 3263 255 63
Analyse cylinder 629/3262: 19%

Linux      0  32 33  3090  9 55  49639424
Linux      0  32 33  3090  9 55  49639424
Linux      0  32 33  3090  9 55  49639424
Linux      0  32 33  3090  9 55  49639424
Linux      0  32 33  3090  9 55  49639424
Linux      0  32 33  3090  9 55  49639424
Linux      0  32 33  3090  9 55  49639424
Linux      0  32 33  3090  9 55  49639424

[Stop]
```

It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from D(eleted) to **L(ogical)**. This way you will be able to recover this partition.

Hint: read [How to recognize primary and logical partitions?](#)

Note: If a partition is listed *(bootable) but if you don't boot from this partition, you can change it to **P**Primary partition.

- Press Enter to proceed.

Partition table recovery

It's now possible to write the new partition structure.

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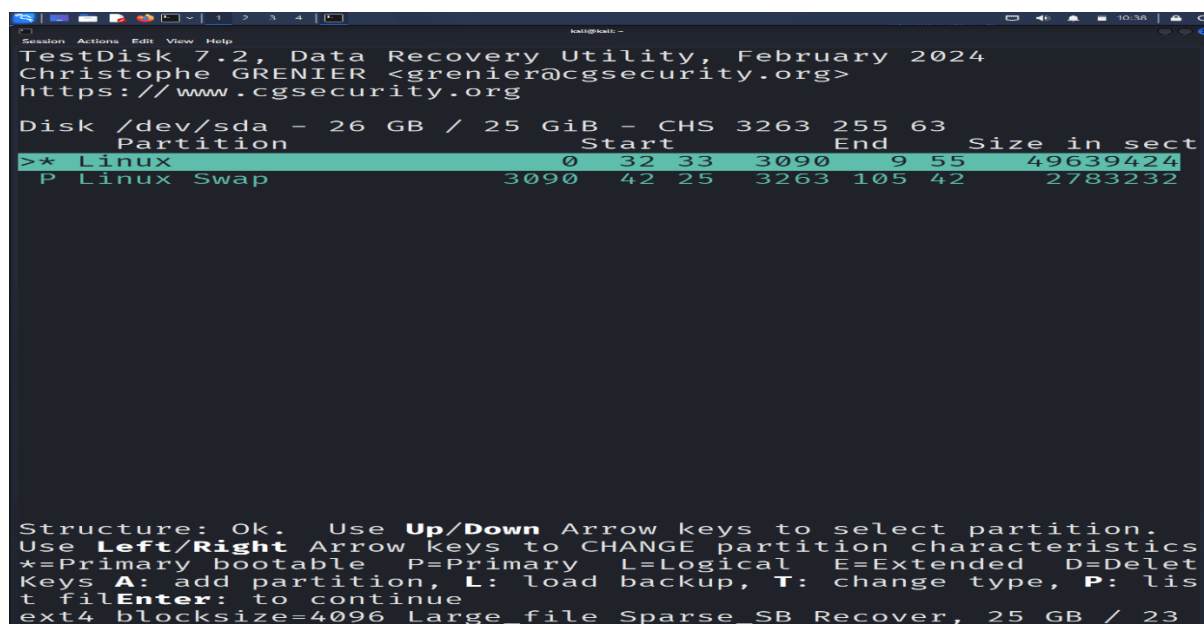
Note: The extended partition is automatically set. TestDisk recognizes this using the different partition structure.

- Press Enter to proceed.

Partition table recovery

It's now possible to write the new partition structure.

Note: The extended partition is automatically set. TestDisk recognizes this using the different partition structure.



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Disk /dev/sda - 26 GB / 25 GiB - CHS 3263 255 63
Partition                                     Start      End      Size in sect
>* Linux                                     0  32 33   3090    9 55   49639424
P Linux Swap                               3090  42 25   3263   105 42   2783232

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics
*=Primary bootable P=Primary L=Logical E=Extended D=Delet
Keys A: add partition, L: load backup, T: change type, P: list
file Enter: to continue
ext4 blocksize=4096 Large_file Sparse_SB Recover, 25 GB / 23
```

- If **all partitions are listed** and only in this case, confirm at **Write** with Enter, y and OK.

Now, the partitions are registered in the partition table.

NTFS Boot sector recovery

The boot sector of the first partition named Partition 1 is still damaged. It's time to fix it. The status of the NTFS boot sector is bad and the backup boot sector is valid. Boot sectors are not identical.

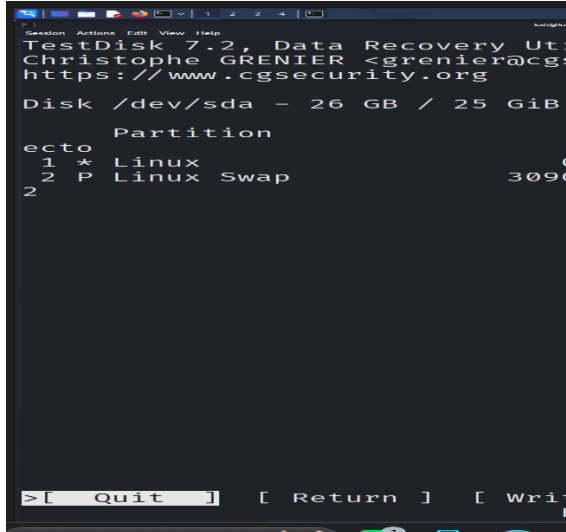
- To copy the backup of the boot sector over the boot sector, select **Backup BS**, validate with Enter, use y to confirm and next OK.

More information about repairing your boot sector under [TestDisk Menu Items](#). The following message is displayed:

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The boot sector and its backup are now both OK and identical: the NTFS boot sector has been successfully recovered.

- Press Enter to quit.



```
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Disk /dev/sda - 26 GB / 25 GiB

Partition
1 * Linux
2 P Linux Swap 3090

> [ Quit ] [ Return ] [ Write ]
```

- TestDisk displays *You have to restart your Computer to access restart your Computer to access your data* so press Enter a last time and reboot your computer



```
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TestDisk will try to restart itself using the sudo command to
get
root (superuser) privileges.

sudo may ask your user password, it doesn't ask for the root
password.
Usually there is no echo or '*' displayed when you type your
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```

The final look of the experiment.